

Before Glossary

A preceding section with substantive text.

Glossary

Term	Definition
AGA	American Gas Association — standards for gas measurement and quality
Agentic AI	AI systems that autonomously plan, execute, and adapt multi-step workflows using external tools and structured protocols
Anti-surge control	Control that maintains margin from compressor surge, commonly by recycling gas; its effectiveness depends on the map, sensors, valve dynamics and implemented logic.
API gravity	Petroleum-liquid density scale: $141.5/SG(60/60\text{ }^{\circ}\text{F}) - 131.5$. The reference temperatures are part of the definition.
Artificial lift	Methods to increase flow from wells, including gas lift, ESP, and rod pumps
Automatic differentiation (AD)	Chain-rule differentiation of implemented elementary operations, accurate to floating-point arithmetic for the differentiable executed path; branches and numerical solvers need appropriate derivative treatment.
Back-pressure	The pressure at the outlet of a production system or equipment item
BHP	Bottomhole pressure at a stated well datum; flowing BHP differs from average reservoir pressure and from shut-in pressure.
BOE	Barrel of oil equivalent; an energy-equivalent reporting quantity whose conversion factor and heating-value basis must be stated.
Bottleneck	The equipment or process constraint that limits total system throughput
Bubble point	For a specified mixture and temperature, the pressure at which an infinitesimal vapor phase first coexists with the liquid; a bubble-point temperature can instead be defined at fixed pressure.
Capacity check	Comparison of operating demands with explicitly configured equipment limits, subject to the rating basis, model applicability and completeness of the checked constraints.
CCR	Cricondenbar: maximum pressure of vapor–liquid coexistence for a specified mixture; not necessarily the mixture critical pressure.
CCT	Cricondetherm: maximum temperature of vapor–liquid coexistence for a specified mixture; not a standard or necessarily the critical temperature.
Choke	A valve that restricts flow, typically at the wellhead or manifold, to control production rate
Compressor curve	Performance map showing the relationship between flow, head, power, and efficiency for a compressor
Compressor surge	Low-flow system instability involving pressure and flow oscillations, potentially including flow reversal; the boundary depends on the compressor and connected system.

Term	Definition
Conformal prediction	Prediction-set calibration with finite-sample marginal coverage under the method's assumptions, commonly exchangeability; it does not guarantee conditional coverage or remain valid under arbitrary distribution shift.
CPA	Cubic Plus Association equation of state
Cv value	Valve flow coefficient based on US gallons per minute of water at about 60 °F producing a 1 psi pressure drop. Gas, cavitation and choked-flow calculations require the appropriate sizing relations.
DeepONet	Deep Operator Network: a learned mapping between function spaces, used as a surrogate for solution operators within a demonstrated training and validation domain.
Dew point	The temperature (at a given pressure) where the first liquid droplet forms from a gas
Digital twin	A computational representation connected to observations of a physical asset, with defined update, synchronization and validation procedures; connection alone does not establish accuracy.
EOS	Equation of state — thermodynamic model relating P, V, T, and composition
ESP	Electrical submersible pump — an artificial lift method
Flash calculation	Equilibrium phase amounts and compositions at specified constraints, such as temperature/pressure or enthalpy/pressure; more than two phases may be present.
Flow assurance	Engineering discipline managing flow-related risks (hydrate, wax, slugging, corrosion)
FNO	Fourier Neural Operator — operator learning architecture that works in Fourier space for efficient PDE surrogate models
FPSO	Floating Production, Storage, and Offloading vessel
GOR	Gas-oil ratio — volume of gas produced per volume of oil at standard conditions
Gymnasium	Python RL library (successor to OpenAI Gym) defining standard environment API: reset(), step(), observation/action/reward spaces
HP separator	High-pressure separator — the first-stage separator in a separation train
HYSYS	Aspen HYSYS, an AspenTech process simulator. Honeywell UniSim Design is a separate product.
IPR	Inflow Performance Relationship — relates wellbore flowing pressure to production rate
ISO 6976	International standard for calculating calorific value of natural gas
JT cooling	Cooling during approximately isenthalpic throttling when the Joule–Thomson coefficient is positive. Throttling can instead heat a fluid when that coefficient is negative.

Term	Definition
K-factor	Here, the Souders–Brown separator capacity factor, with velocity units; distinguish it from the dimensionless phase-equilibrium ratio $K_i = y_i/x_i$.
LP separator	Low-pressure separator — a later stage in a separation train
LMTD	Log-mean temperature difference — used in heat exchanger design
MCP	Model Context Protocol — standardized protocol enabling LLMs to discover and use external tools (e.g., simulation capabilities)
MEG	Mono-ethylene glycol — hydrate inhibitor
Mixing rule	Method for combining pure-component EOS parameters to model mixtures
MMSCFD	Million standard cubic feet per day — gas flow rate unit
Monte Carlo dropout	Approximate uncertainty estimation from repeated stochastic network evaluations with dropout; prediction spread needs calibration and is not automatically a reliable physical-error bound.
Multiphase flow	Simultaneous transport of two or more phases, for example gas–liquid, oil–water or gas–oil–water.
NeqSim	Non-equilibrium Simulator — open-source Java library for thermodynamic and process simulation
NGL	Natural gas liquids — ethane, propane, butane, and heavier hydrocarbons
NORSOK	Norwegian standards for petroleum industry activities
NPV	Net present value: the sum of cash flows, including initial investment, discounted to a specified valuation date using a stated discount-rate convention.
OPR	Outflow performance relationship: pressure required to transport a specified rate through tubing and downstream equipment for stated outlet and thermal conditions.
Phase envelope	Diagram showing the boundary between single-phase and two-phase regions
PI	Productivity index: production rate divided by pressure drawdown on a stated flow/pressure basis; the linear relation has an applicability range.
Polytropic efficiency	For compression, the ratio of differential ideal pressure work to differential actual enthalpy rise along the compression path; real-gas evaluation requires thermodynamic properties and a defined method.
PR EOS	Peng–Robinson equation of state
ProcessSystem	NeqSim Java class representing a complete process flowsheet
PVT	Pressure–Volume–Temperature — laboratory measurements of fluid properties
Riser	Vertical or near-vertical pipe connecting subsea infrastructure to a surface facility

Term	Definition
SRK EOS	Soave-Redlich-Kwong equation of state
Separator	Vessel that separates a multiphase stream into gas, oil, and water phases
Slug flow	Intermittent multiphase flow pattern with alternating gas and liquid slugs
Surge line	The locus of minimum stable flow points on a compressor performance map
TEG	Triethylene glycol — used for gas dehydration
TPflash	Temperature-pressure flash — phase equilibrium calculation at given T and P
Turndown	The ability of equipment to operate below design capacity
UA	Overall heat transfer coefficient times area — characterizes heat exchanger performance
Utilization	Dimensionless operating demand divided by the relevant declared limit; separate limits may apply to power, actual volume, gas load, liquid retention or other quantities.
VFP	Vertical Flow Performance — well/tubing pressure-rate relationship tables
VLE	Vapor-liquid equilibrium
Wellhead	Surface equipment at the top of a well
Affinity laws	Approximate centrifugal-machine similarity relations: flow scales with speed, head with speed squared and power with density times speed cubed, for fixed geometry and comparable operating conditions.
Brayton cycle	Thermodynamic cycle underlying gas turbine operation — compression, combustion, expansion
CCGT	Combined cycle gas turbine — gas turbine with heat recovery steam generator and steam turbine
CapacityConstrainedEquipment	NeqSim interface enabling equipment to self-report capacity limits (HARD, SOFT, DESIGN types)
CAPEX	Capital expenditure — investment cost for equipment, facilities, or modifications
Carbon intensity	Emissions divided by a declared production or energy measure, with system boundary, gas coverage and CO ₂ -equivalent convention stated.
Decline curve	Empirical production-rate trend, such as an Arps model; fitted coefficients and flow-regime applicability do not by themselves establish recoverable reserves.
EUR	Estimated ultimate recovery — total hydrocarbon volume expected from a well or field
HRSG	Heat recovery steam generator — extracts energy from gas turbine exhaust

Term	Definition
HVDC	High-voltage direct current — used for long-distance power transmission to offshore platforms
IRR	Discount rate at which project NPV is zero; cash-flow patterns can produce multiple IRRs or no economically useful root.
KHI	Kinetic hydrate inhibitor — polymer that slows hydrate nucleation and growth
Latin Hypercube Sampling	Stratified sampling that covers each marginal probability interval once; it does not guarantee uniform coverage of all joint combinations.
LoopedPipeNetwork	NeqSim class for solving interconnected well/pipeline network hydraulics
MPC	Model predictive control — advanced control strategy using a process model for future prediction
Monte Carlo simulation	Computational method using random sampling to model uncertainty in outcomes
MultiScenarioVFPGenerator	NeqSim class for generating VFP tables across parameter ranges (WC, GOR, pressure)
OPEX	Operating expenditure — recurring costs for running a facility
OptimizationConfig	NeqSim builder class for configuring search algorithms, tolerances, and caching
ONNX	Open Neural Network Exchange — standardized format for exporting trained neural network models between frameworks (PyTorch, TensorFlow, Java)
OptimizationResult	NeqSim class containing optimal rate, bottleneck, iteration history, and diagnostics
ORC	Organic Rankine Cycle — power generation from low-grade waste heat using organic working fluids
P10/P50/P90	Quantile labels requiring an explicit convention. Under non-exceedance, P10 is the 10th percentile; petroleum resource exceedance notation commonly uses P10 for the high case and P90 for the low case.
Pareto front	Set of solutions where no objective can be improved without worsening another
PFS	Power from shore — supplying offshore platforms with onshore electricity
PINN	Physics-informed neural network: includes equation residuals or physical constraints in training. Soft penalty terms encourage, but do not guarantee, conservation or an accurate solution.
PPO	Proximal Policy Optimization — policy gradient RL algorithm using clipped surrogate objective for stable training
ProcessAutomation	NeqSim API providing string-addressable variable access for process models

Term	Definition
ProcessModel	NeqSim class for composing multiple ProcessSystem objects into a multi-area plant model
ProductionOptimizer	NeqSim class for automated production rate maximization subject to equipment constraints
Rachford-Rice	Equation relating phase fraction to K-values and feed composition in flash calculations
Recycle	NeqSim recycle-stream convergence equipment supporting direct substitution and optional acceleration; acceptance still requires residual and conservation checks.
Reinforcement learning (RL)	Training an agent to make sequential decisions by interacting with an environment and maximizing cumulative reward
Robust optimization	Optimization imposing feasibility or performance requirements over a specified uncertainty set; coverage is limited to that set and the chosen formulation.
SAC	Soft Actor-Critic — off-policy RL algorithm combining actor-critic with maximum entropy for exploration
Scope 1/2 emissions	Scope 1 covers direct emissions from owned or controlled sources; Scope 2 covers indirect emissions from purchased electricity, steam, heat and cooling.
Sim-to-real transfer	Deploying policies trained in simulation to real-world systems, addressing the gap between model and reality
Stochastic optimization	Optimization using probability models for uncertain inputs, through sampling, probability-weighted scenarios or other stochastic formulations.
Surrogate model	Simplified mathematical model (e.g., neural network) trained to approximate a complex simulator
SurrogateEquipment	Proposed NeqSim class wrapping a trained neural network in the standard equipment interface for hybrid physics-AI simulation
Tornado diagram	Ranked one-at-a-time sensitivity ranges for a declared input variation and baseline; the bars are not generally a variance decomposition or probability distribution.
UQ	Uncertainty quantification: identifying, propagating and assessing uncertainties in inputs, observations, parameters and models with explicit assumptions and calibration evidence.
VFD	Variable frequency drive — controls motor speed for energy-efficient equipment operation
Wobbe index	Calorific value divided by the square root of gas relative density to air, using consistent reference conditions and gross/net basis; this is an interchangeability indicator, not a complete burner-compatibility test.